



Diffusion Model

Presentation

Trịnh Ngọc Huỳnh - Đặng Xuân Lộc



Table of Contents

01

Introduction

02

Backgrounds

03

**Generative
Model**

04

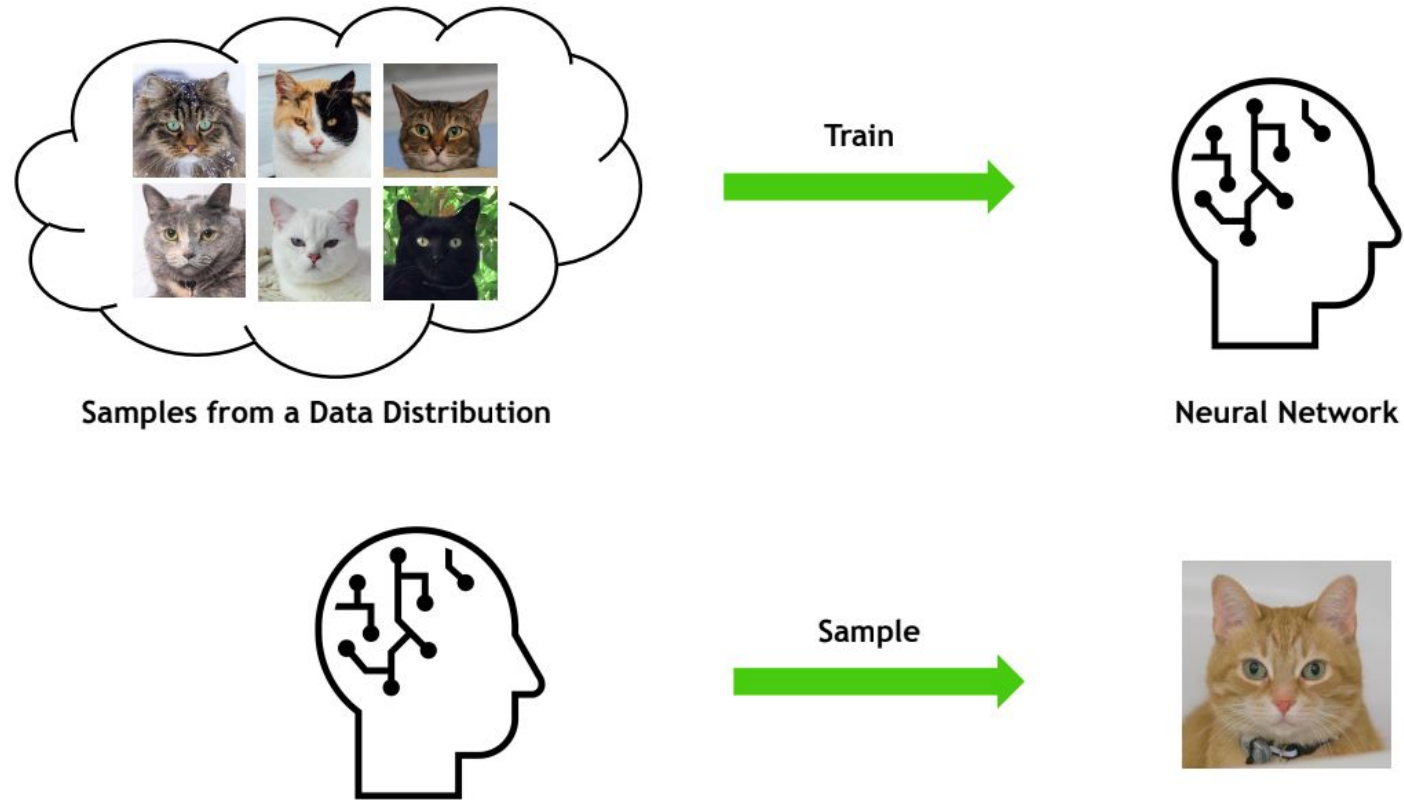
**Diffusion
Model**

05

Demo

01 | Introduction

Deep Generative Learning Learning to generate data



01 | Introduction

Text-to-Image Generation

DALL·E 2

"a teddy bear on a skateboard in times square"



["Hierarchical Text-Conditional Image Generation with CLIP Latents"](#)
Ramesh et al., 2022

Imagen

A group of teddy bears in suit in a corporate office celebrating the birthday of their friend. There is a pizza cake on the desk.



["Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding"](#), Saharia et al., 2022

Supervised vs Unsupervised Learning

Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn a *function* to map $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, image captioning, etc.



→ Cat

Classification

Supervised vs Unsupervised Learning

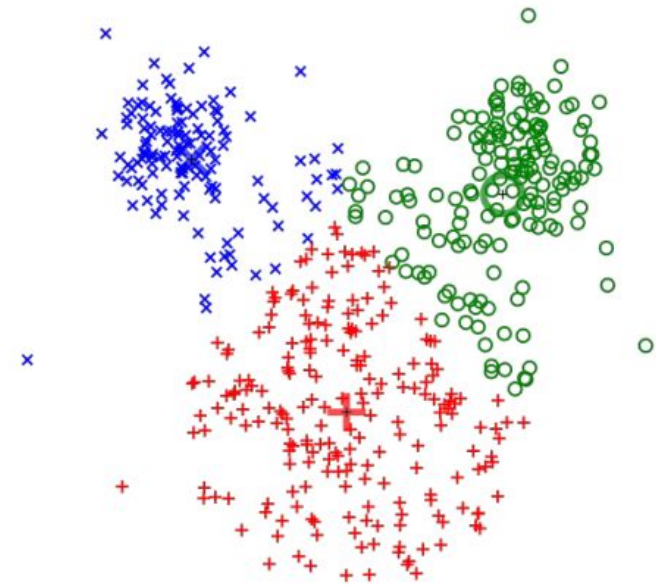
Unsupervised Learning

Data: x

Just data, no labels!

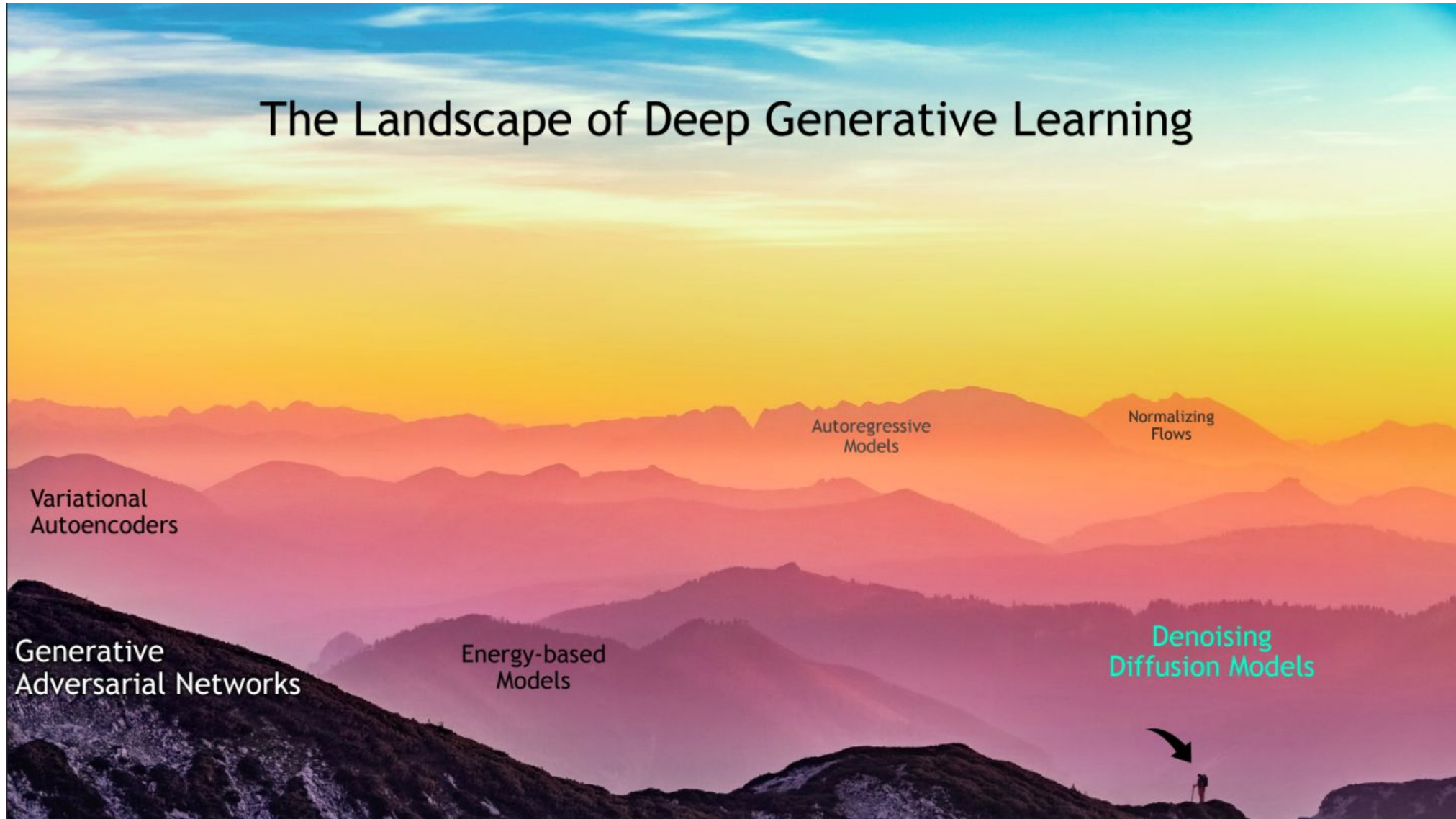
Goal: Learn some underlying hidden *structure* of the data

Examples: Clustering, dimensionality reduction, density estimation, etc.



K-means clustering

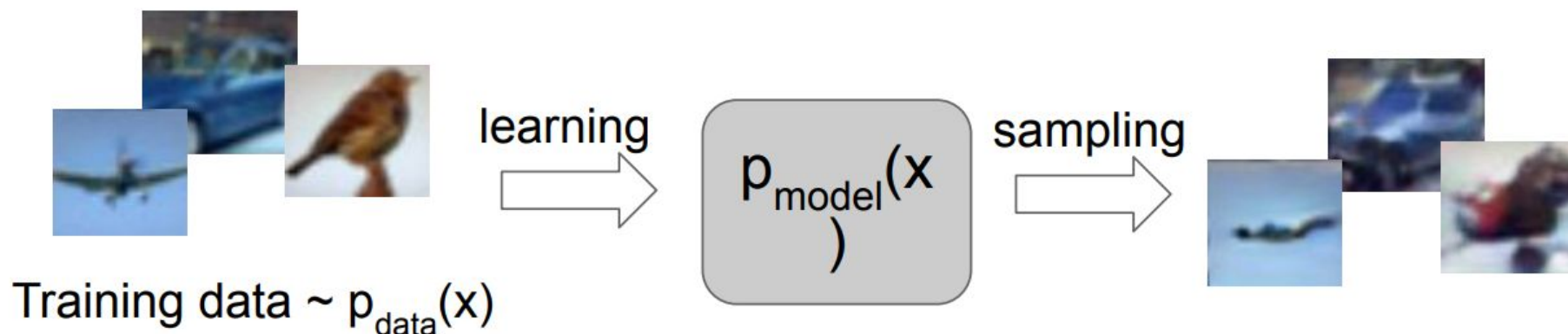
03 | Generative Model



03 | Generative Model

Generative Modeling

Given training data, generate new samples from same distribution



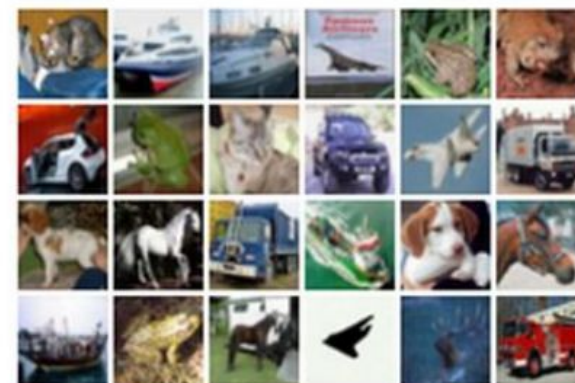
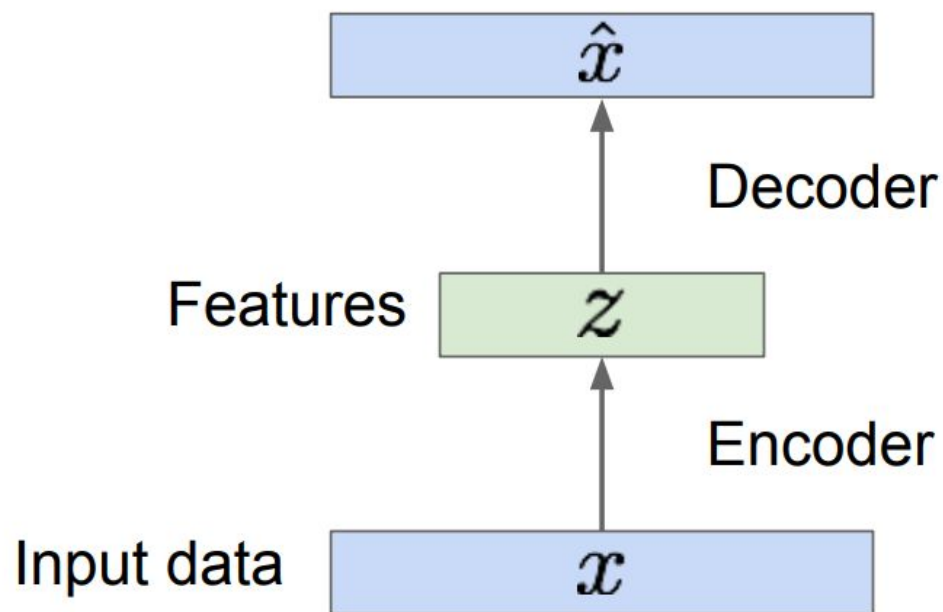
Objectives:

1. Learn $p_{\text{model}}(x)$ that approximates $p_{\text{data}}(x)$
2. **Sampling new x from $p_{\text{model}}(x)$**

03 | Generative Model: VAE

Some background first: Autoencoders

Unsupervised approach for learning a lower-dimensional feature representation from unlabeled training data



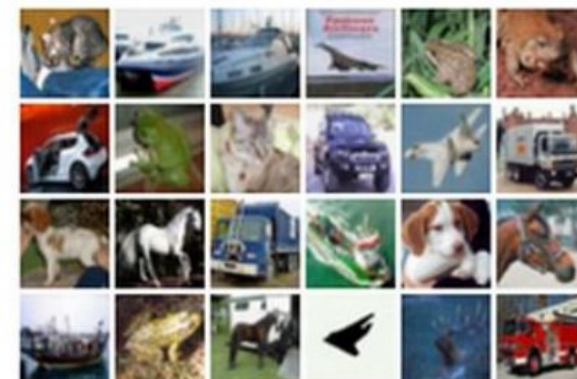
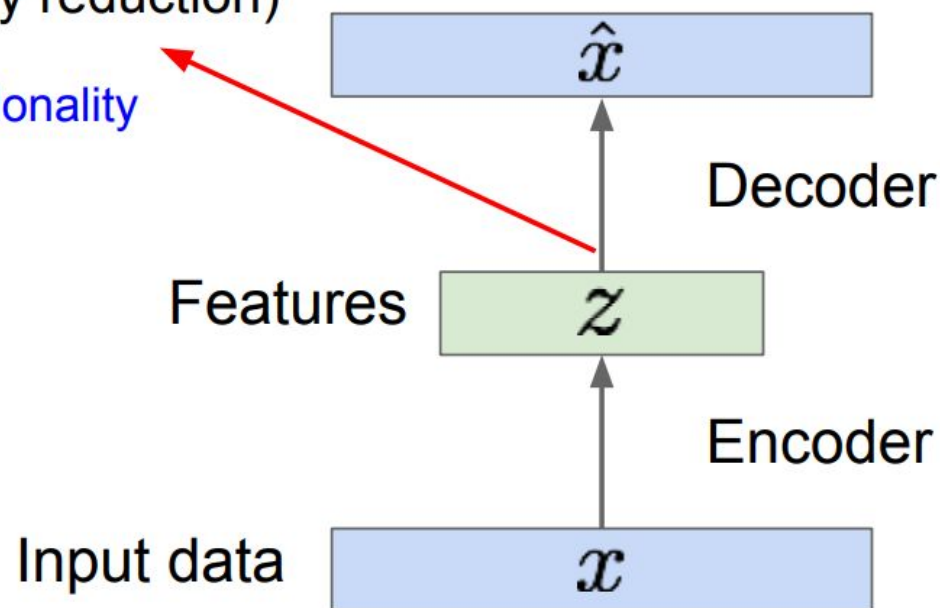
03 | Generative Model: VAE

Some background first: Autoencoders

Unsupervised approach for learning a lower-dimensional feature representation from unlabeled training data

$\mathbf{z}$ usually smaller than $\mathbf{x}$
(dimensionality reduction)

Q: Why dimensionality reduction?



03 | Generative Model: VAE

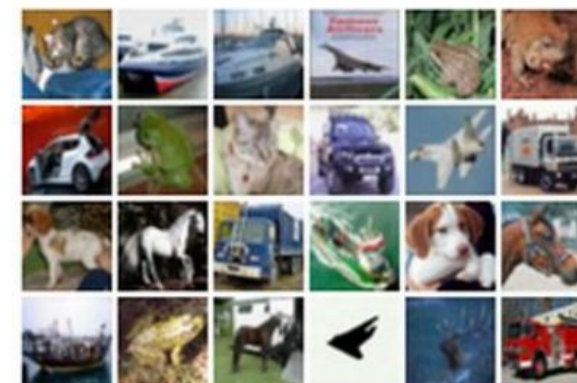
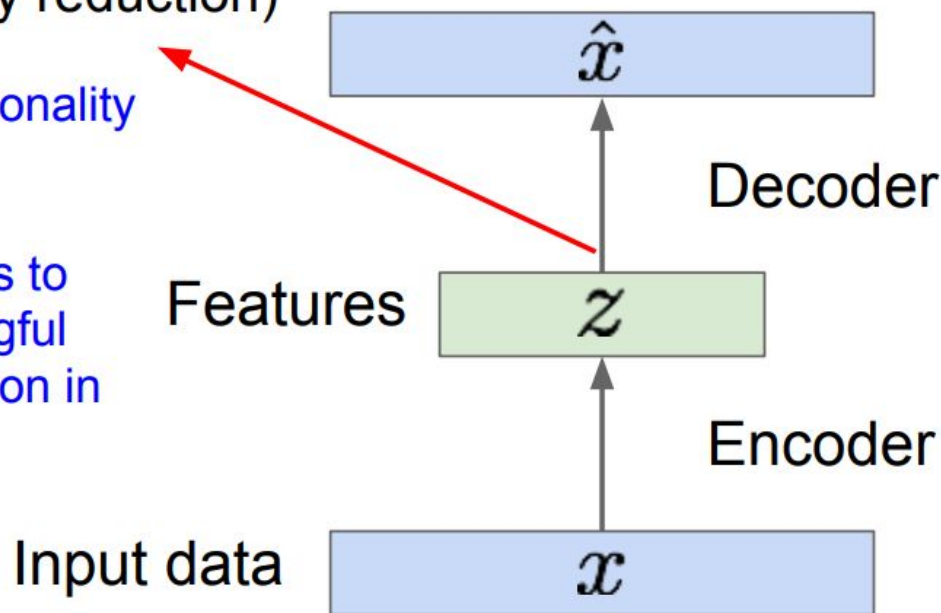
Some background first: Autoencoders

Unsupervised approach for learning a lower-dimensional feature representation from unlabeled training data

$\mathbf{z}$ usually smaller than $\mathbf{x}$
(dimensionality reduction)

Q: Why dimensionality reduction?

A: Want features to capture meaningful factors of variation in data

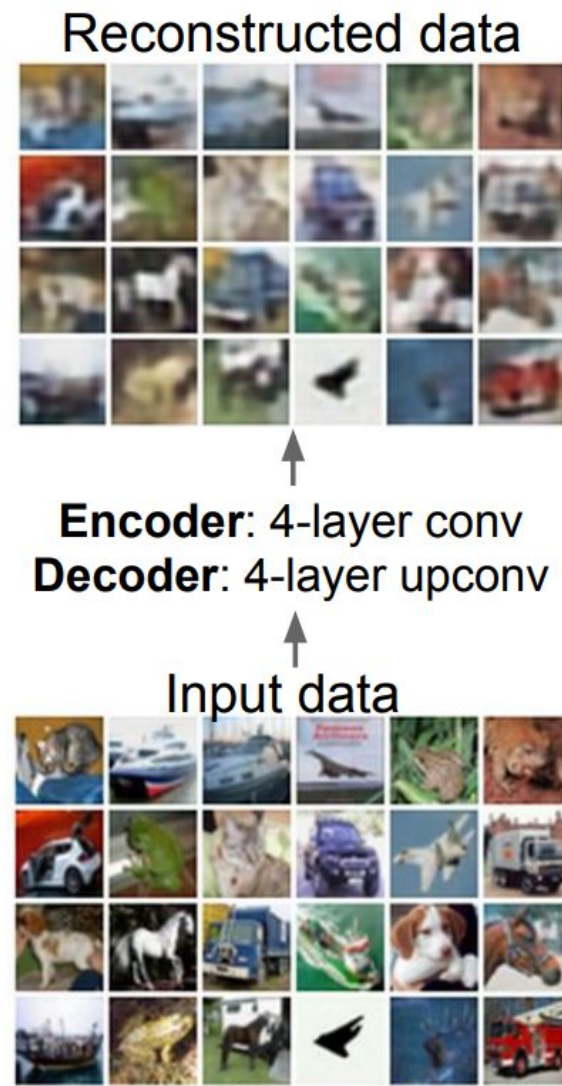
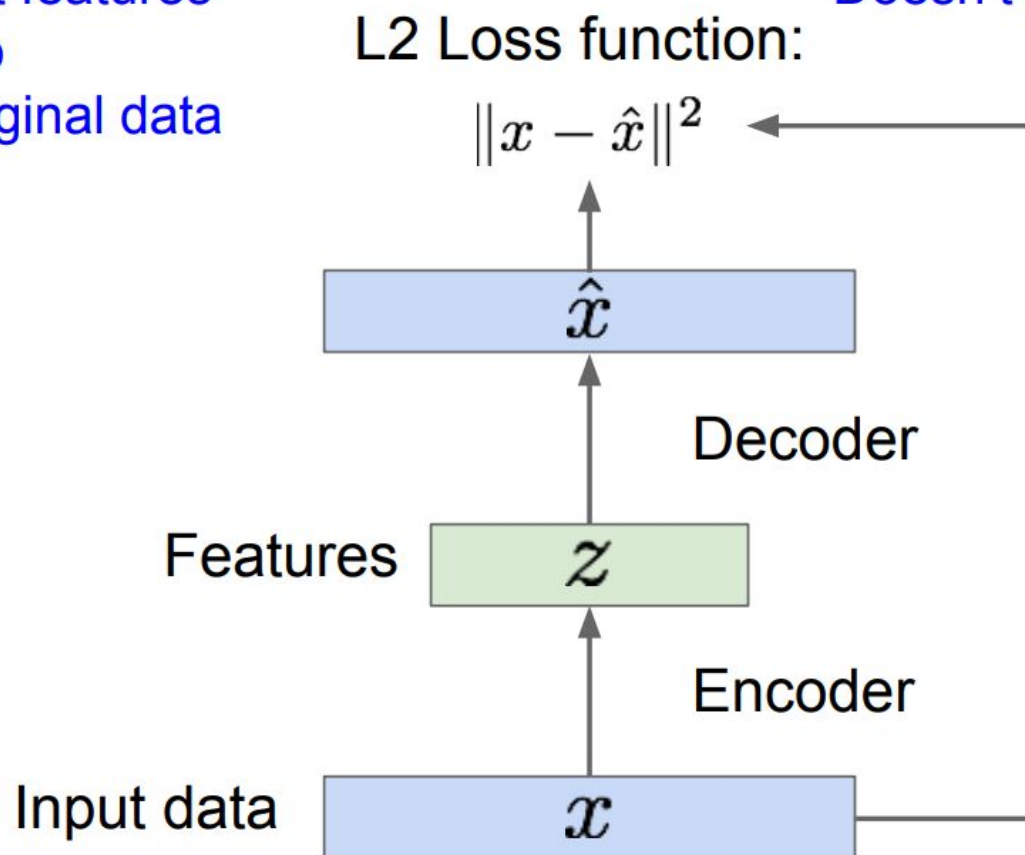


03 | Generative Model: VAE

Some background first: Autoencoders

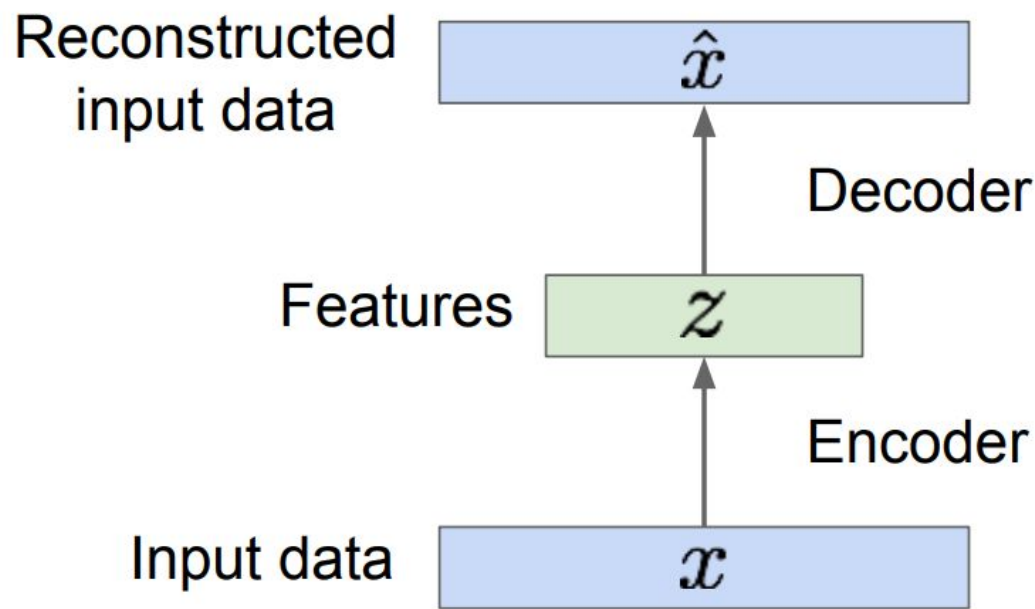
Train such that features
can be used to
reconstruct original data

Doesn't use labels!



03 | Generative Model: VAE

Some background first: Autoencoders



Autoencoders can reconstruct data, and can learn features to initialize a supervised model

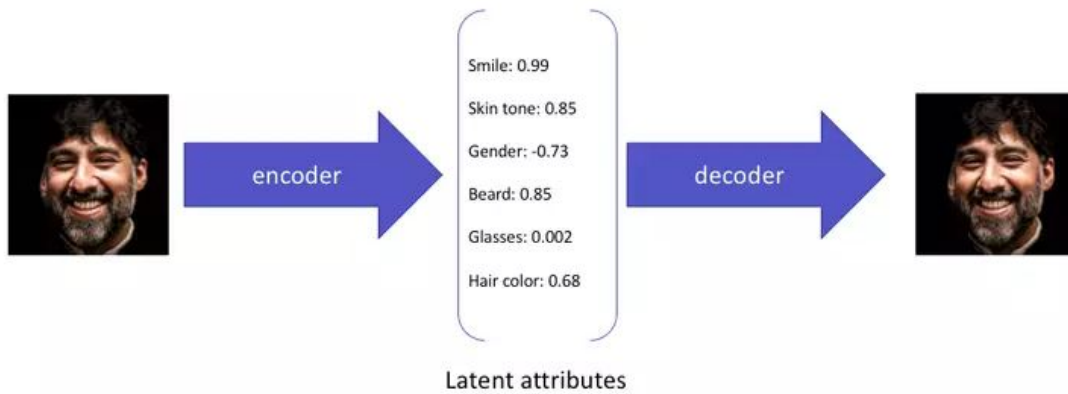
Features capture factors of variation in training data.

But we can't generate new images from an autoencoder because we don't know the space of z .

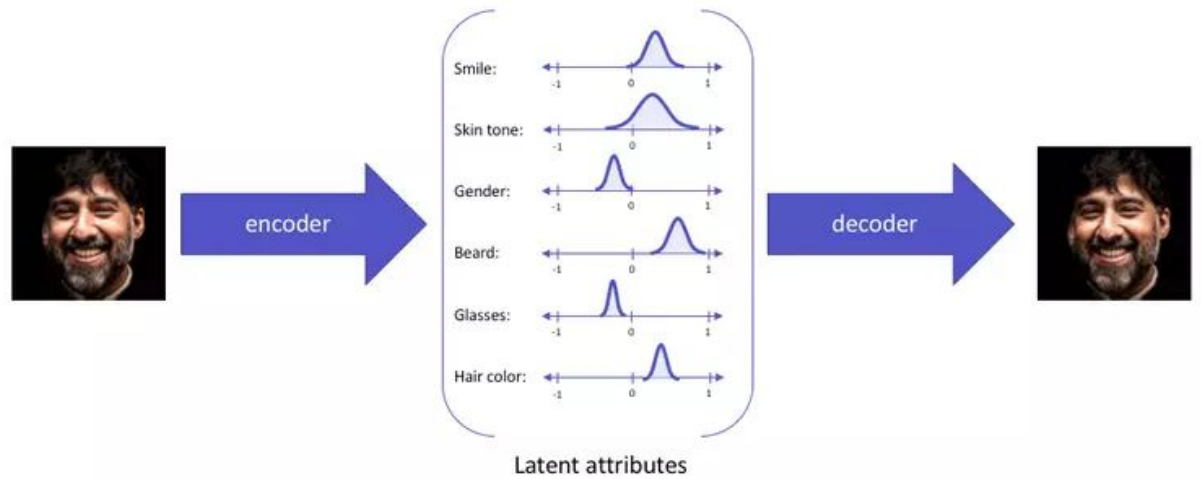
How do we make autoencoder a **generative model**?

03 | Generative Model: VAE

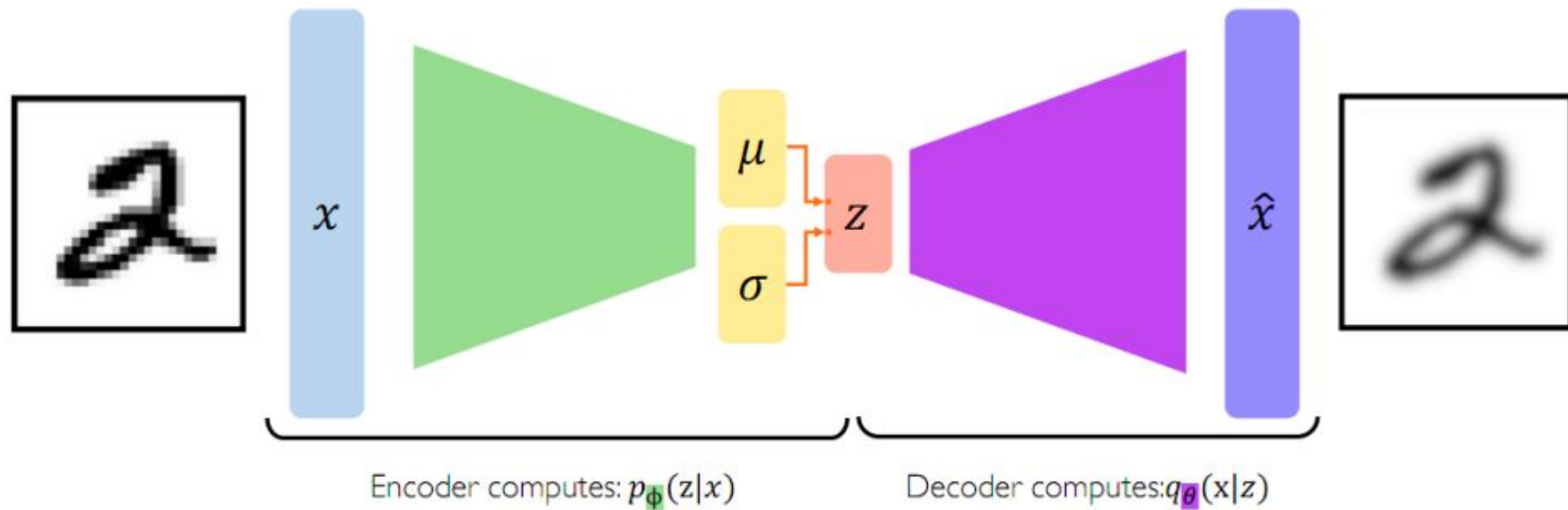
AutoEncoder



Variational Auto Encoder

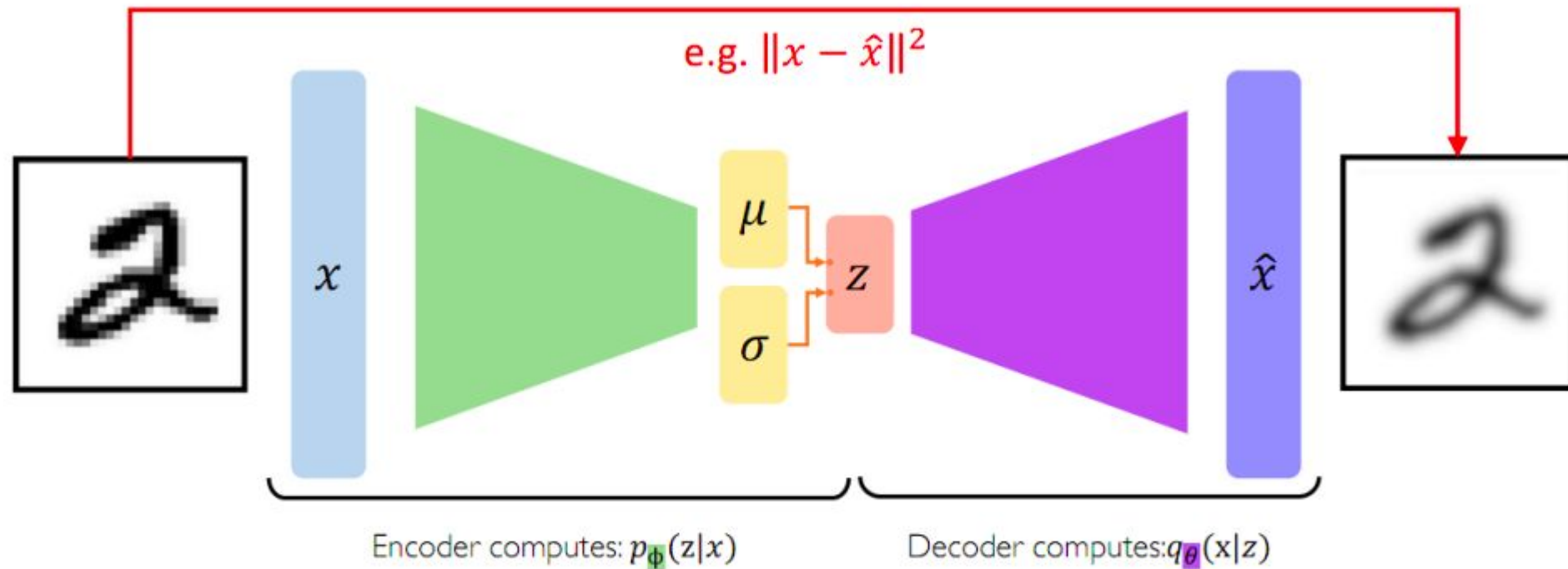


03 | Generative Model: VAE



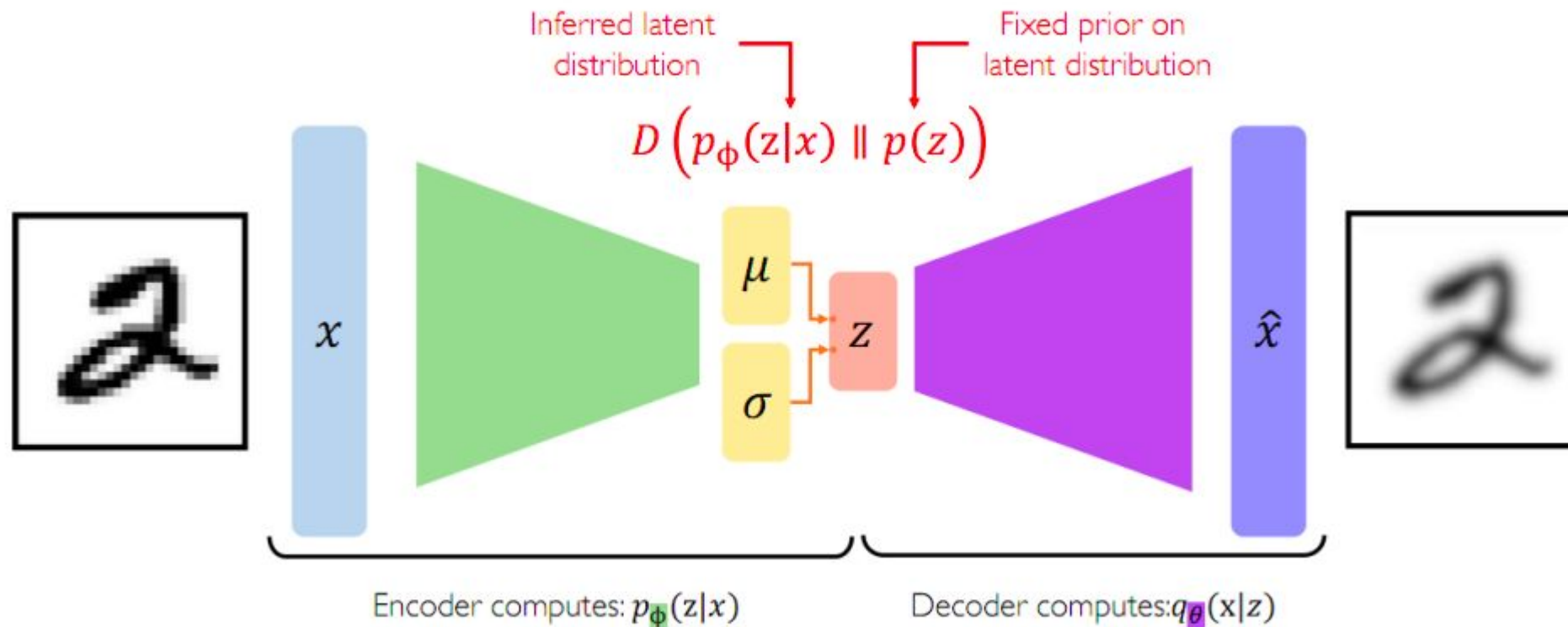
$$\mathcal{L}(\phi, \theta, x) = (\text{reconstruction loss}) + (\text{regularization term})$$

03 | Generative Model: VAE



$$\mathcal{L}(\phi, \theta, x) = \boxed{\text{reconstruction loss}} + (\text{regularization term})$$

03 | Generative Model: VAE

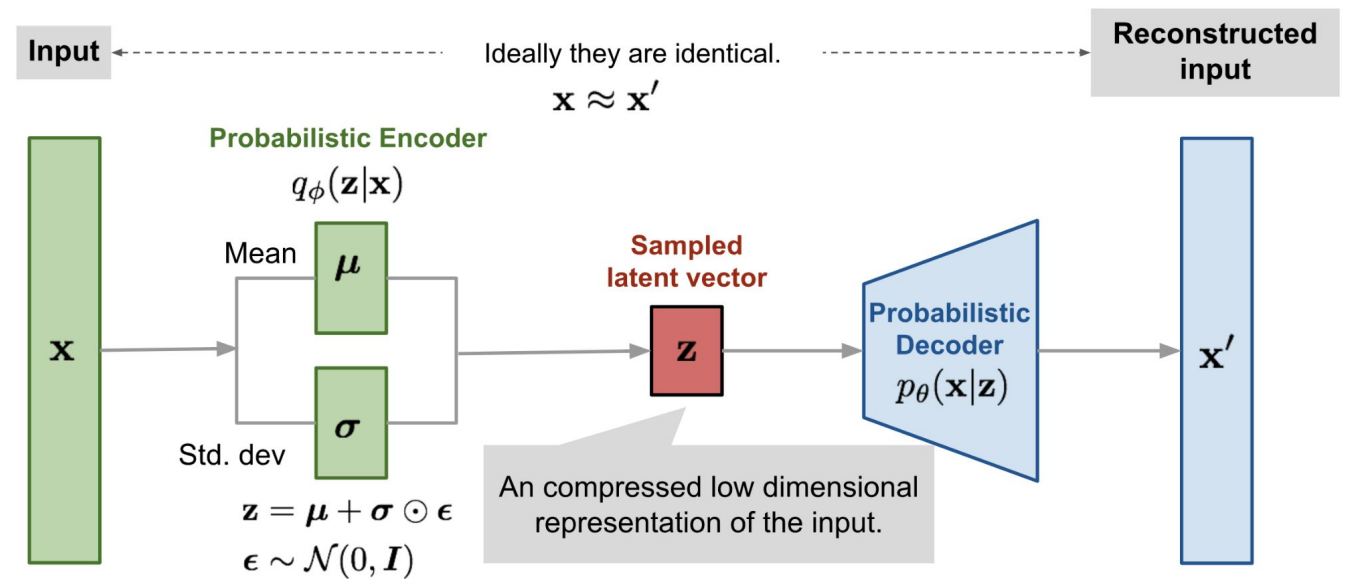


$$\mathcal{L}(\phi, \theta, x) = (\text{reconstruction loss}) + (\text{regularization term})$$

03 | Generative Model: VAE

VAE - Variational Autoencoder

- Blur sample
- Unrealistic output



03 | Generative Model: GAN

Problem: Want to sample from complex, high-dimensional training distribution. No direct way to do this!

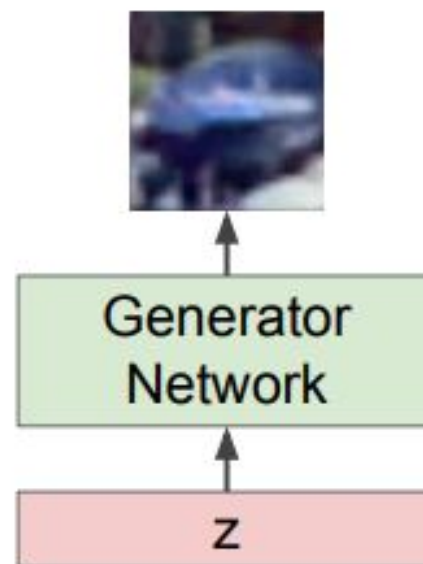
Solution: Sample from a simple distribution we can easily sample from, e.g. random noise. Learn transformation to training distribution.

Q: What can we use to represent this complex transformation?

A: A neural network!

Output: Sample from training distribution

Input: Random noise



03 | Generative Model: GAN

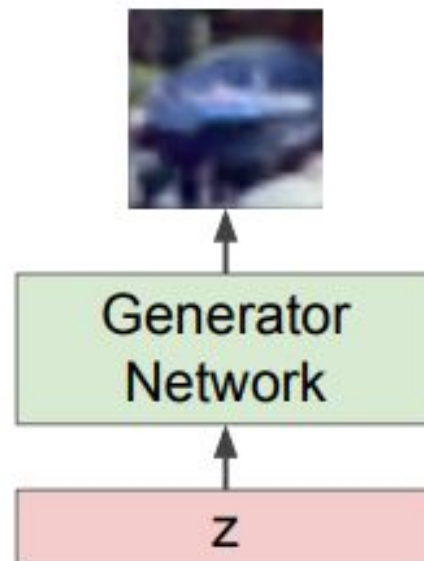
Problem: Want to sample from complex, high-dimensional training distribution. No direct way to do this!

Solution: Sample from a simple distribution we can easily sample from, e.g. random noise. Learn transformation to training distribution.

But we don't know which sample z maps to which training image $\rightarrow$ can't learn by reconstructing training images

Output: Sample from training distribution

Input: Random noise



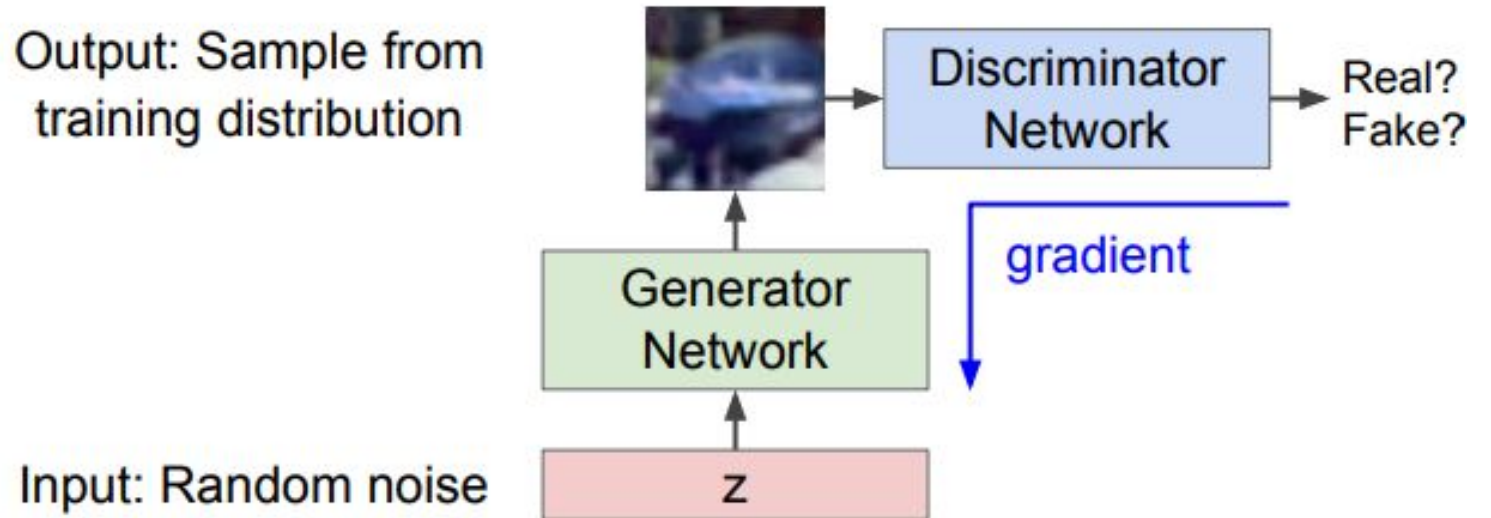
03 | Generative Model: GAN

Problem: Want to sample from complex, high-dimensional training distribution. No direct way to do this!

Solution: Sample from a simple distribution we can easily sample from, e.g. random noise. Learn transformation to training distribution.

But we don't know which sample z maps to which training image -> can't learn by reconstructing training images

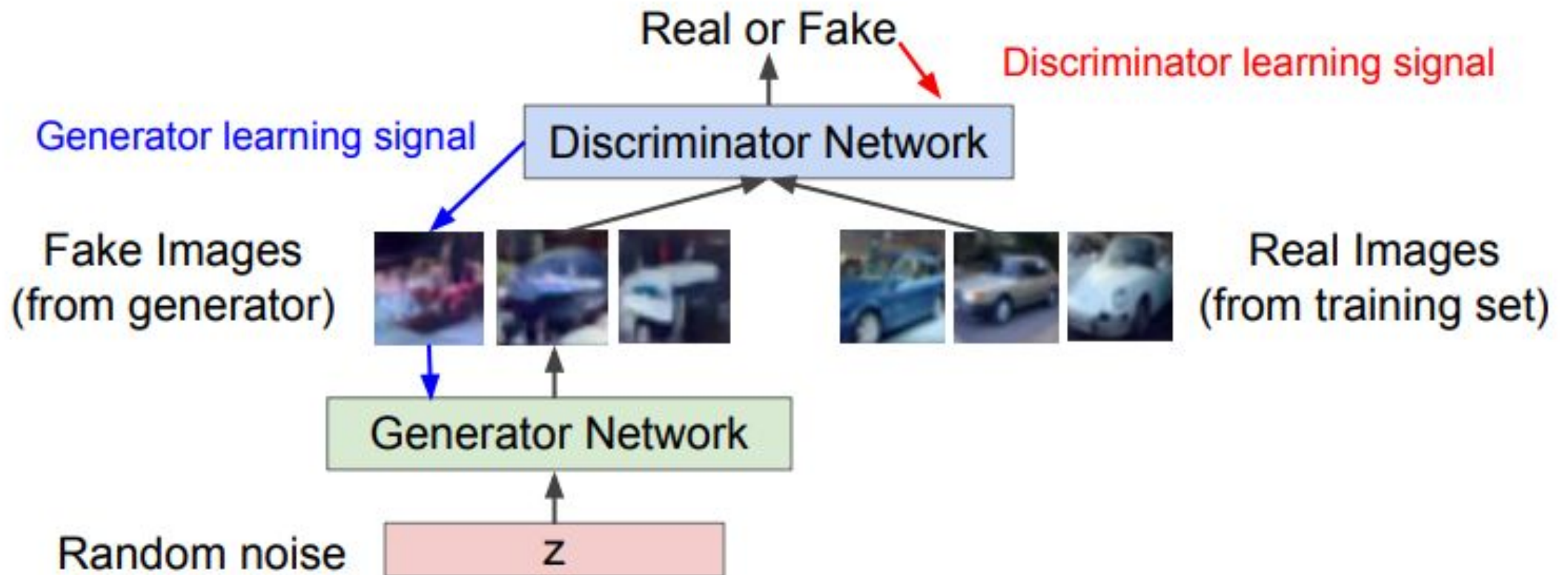
Solution: Use a discriminator network to tell whether the generated image is within data distribution or not



03 | Generative Model: GAN

Discriminator network: try to distinguish between real and fake images

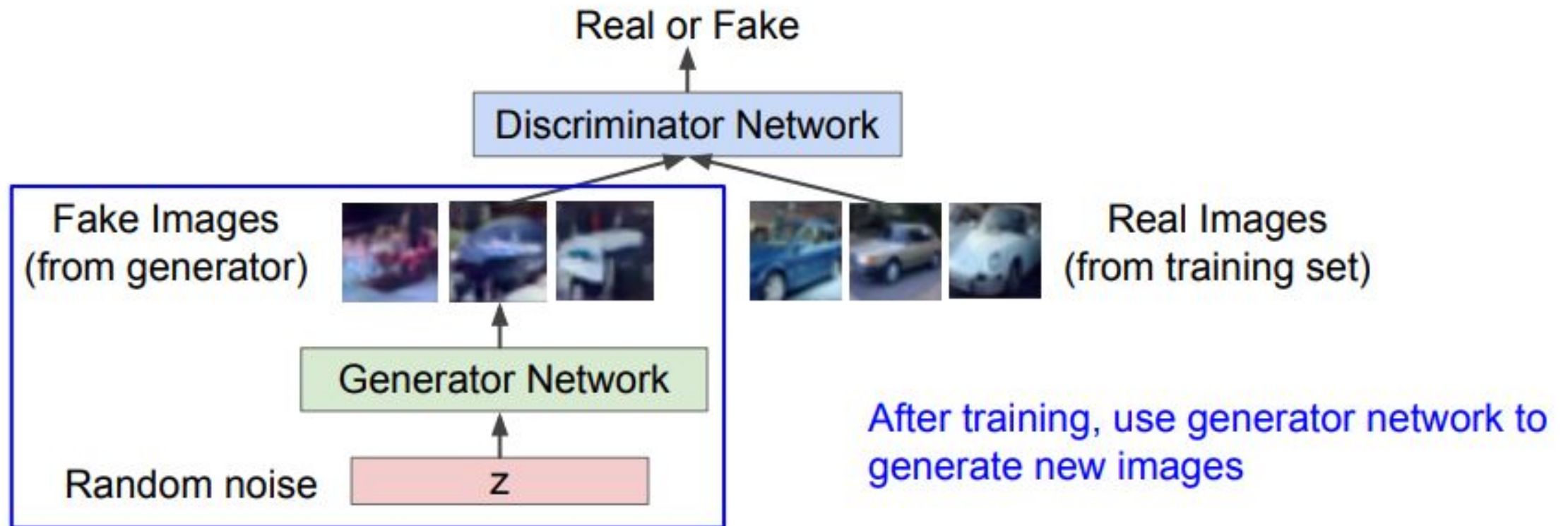
Generator network: try to fool the discriminator by generating real-looking images



03 | Generative Model: GAN

Generator network: try to fool the discriminator by generating real-looking images

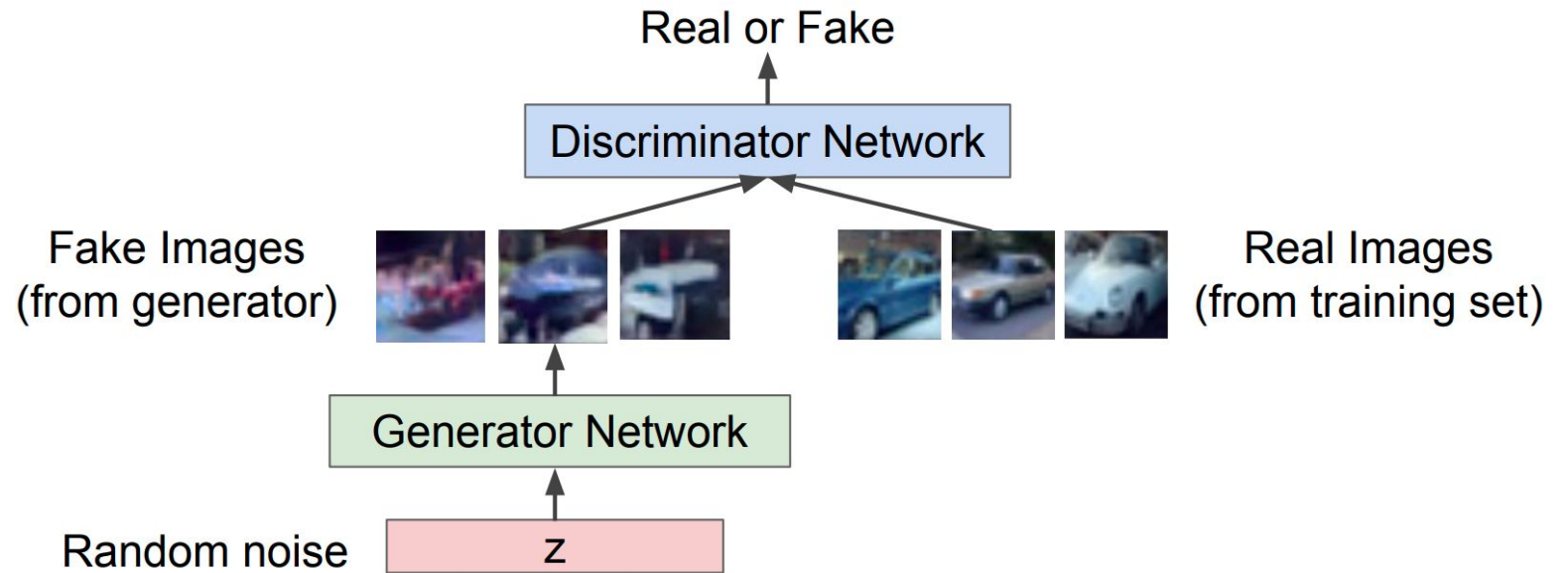
Discriminator network: try to distinguish between real and fake images



03 | Generative Model: GAN

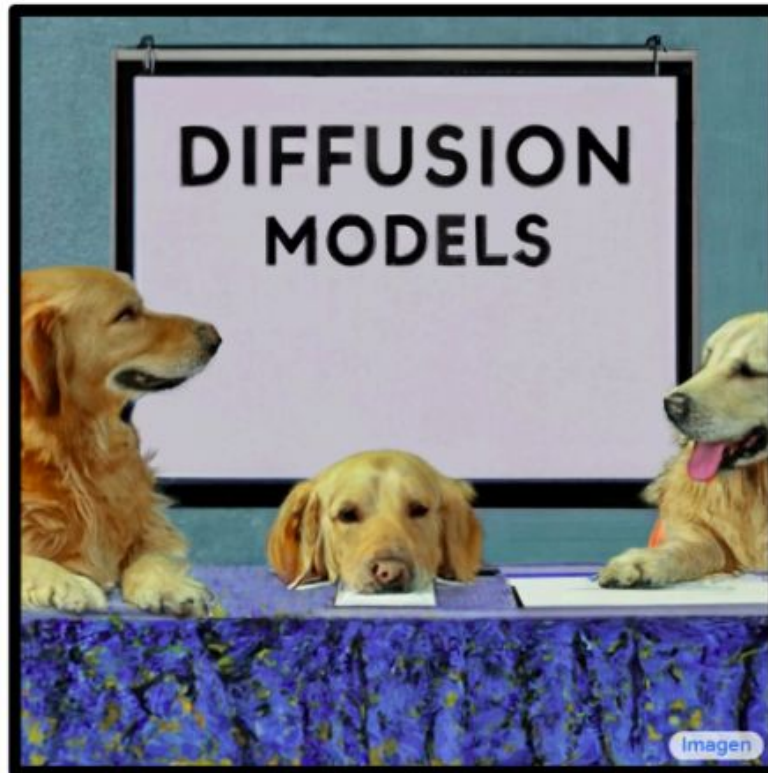
GAN - Generative Adversarial Network

- Unstable Training
- Model collapse
- Diminished gradient



04 | Diffusion Model

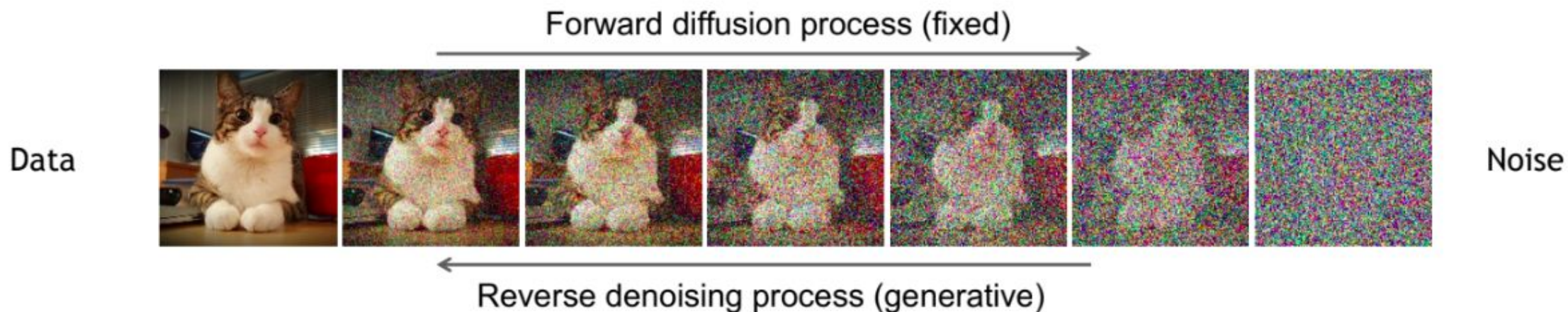
Denoising Diffusion Probabilistic Models



04 | Diffusion Model

Denoising diffusion models consist of two processes:

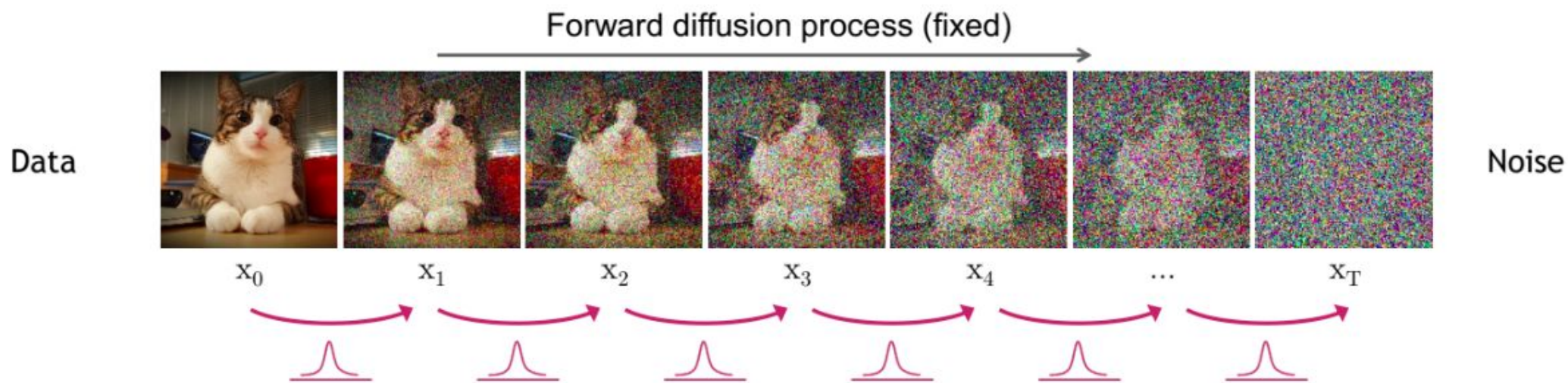
- Forward diffusion process that gradually adds noise to input
- Reverse denoising process that learns to generate data by denoising



04 | Diffusion Model

Forward Diffusion Process

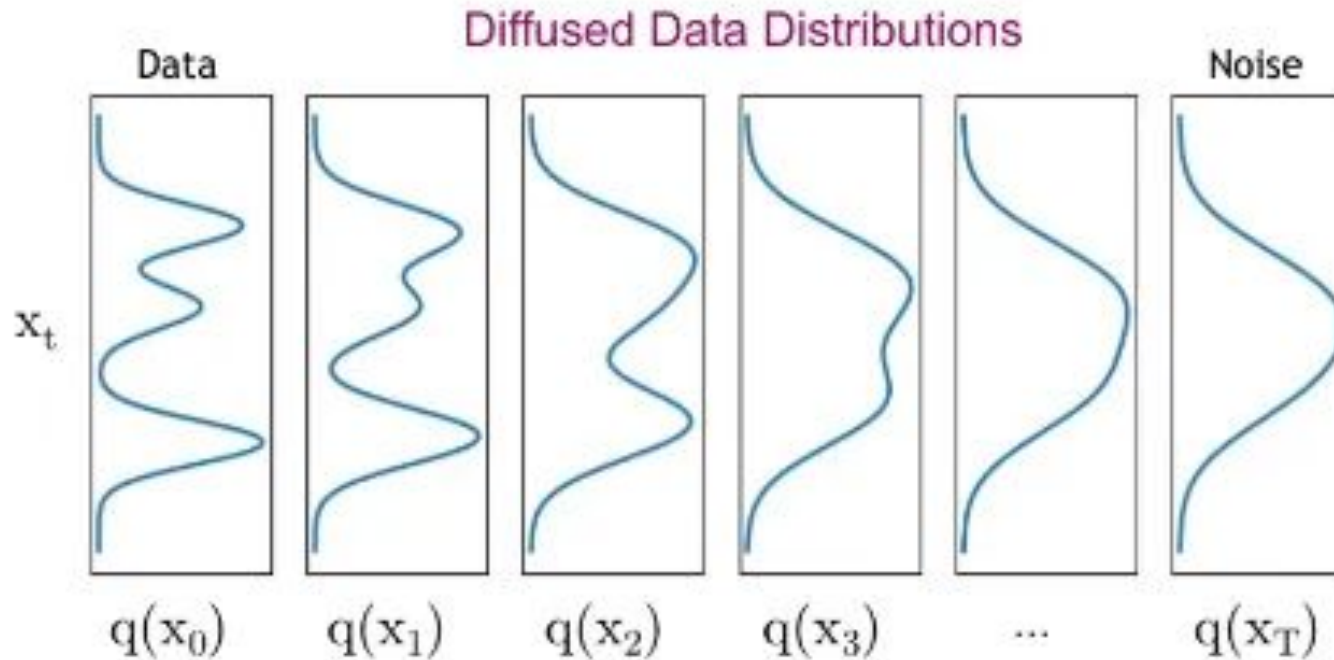
The formal definition of the forward process in T steps:



$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}) \quad \rightarrow \quad q(\mathbf{x}_{1:T} | \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1}) \quad (\text{joint})$$

04 | Diffusion Model

What happen to a distribution in the forward diffusion



$$\mathbf{x}_0 \sim \mathcal{P}$$

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \epsilon \sim q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t I)$$

$$\beta_t \in (0, 1) \rightarrow \mathbf{x}_t | \mathbf{x}_0 \xrightarrow{\mathcal{D}} \mathcal{N}(0, I)$$

$$T \text{ đủ lớn } (1000 \rightarrow 4000) : \mathbf{x}_t \sim \mathcal{N}(0, I)$$

04 | Diffusion Model

Forward Diffusion Process

A nice property of the above process is that we can sample $\mathbf{x}_t$ at any arbitrary time step t

Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$:

$$\begin{aligned}\mathbf{x}_t &= \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} && \text{; where } \boldsymbol{\epsilon}_{t-1}, \boldsymbol{\epsilon}_{t-2}, \dots \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \\ &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\boldsymbol{\epsilon}}_{t-2} && \text{; where } \bar{\boldsymbol{\epsilon}}_{t-2} \text{ merges two Gaussians (*)}. \\ &= \dots \\ &= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}\end{aligned}$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

04 | Diffusion Model

Reverse Denoising Process

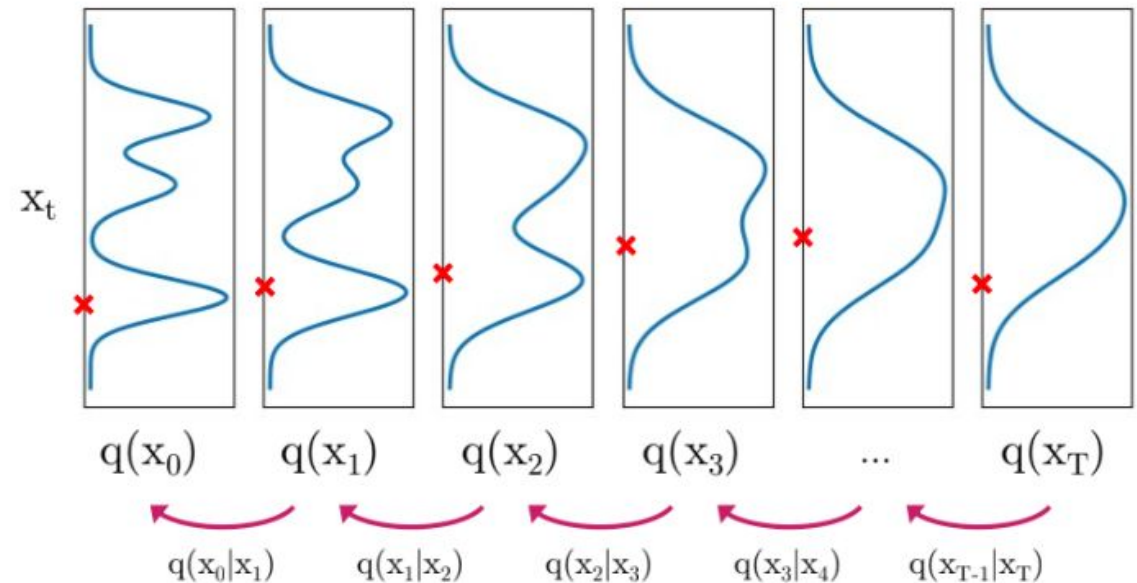
Recall, that the diffusion parameters are designed such that $q(\mathbf{x}_T) \approx \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$

Generation:

Sample $\mathbf{x}_T \sim \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$

Iteratively sample $\mathbf{x}_{t-1} \sim \underbrace{q(\mathbf{x}_{t-1}|\mathbf{x}_t)}_{\text{True Denoising Dist.}}$

Diffused Data Distributions



04 | Diffusion Model

Reverse Denoising Process

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) \sim q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\boldsymbol{\beta}}_t \mathbf{I})$$

04 | Diffusion Model

Reverse Denoising Process

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) \sim q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\boldsymbol{\beta}}_t \mathbf{I})$$

$$q(x_{t-1}|x_t, x_0) = \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

04 | Diffusion Model

Reverse Denoising Process

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) \sim q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\boldsymbol{\beta}}_t \mathbf{I})$$

$$q(x_{t-1}|x_t, x_0) = \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) \quad \tilde{\boldsymbol{\beta}}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \boldsymbol{\beta}_t$$

04 | Diffusion Model

Reverse Denoising Process

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) \sim q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\boldsymbol{\beta}}_t \mathbf{I})$$

$$q(x_{t-1}|x_t, x_0) = \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

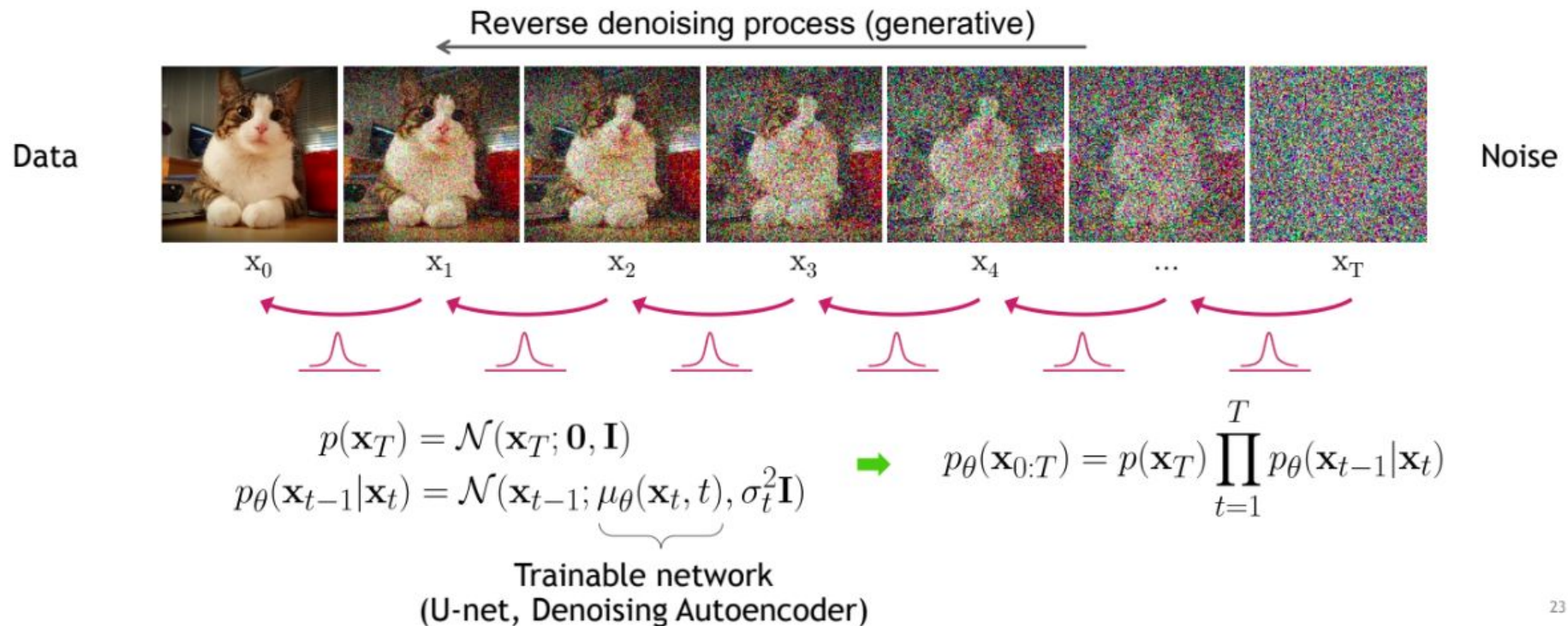
$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) \quad \tilde{\boldsymbol{\beta}}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \boldsymbol{\beta}_t$$

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \boldsymbol{\epsilon} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$$

04 | Diffusion Model

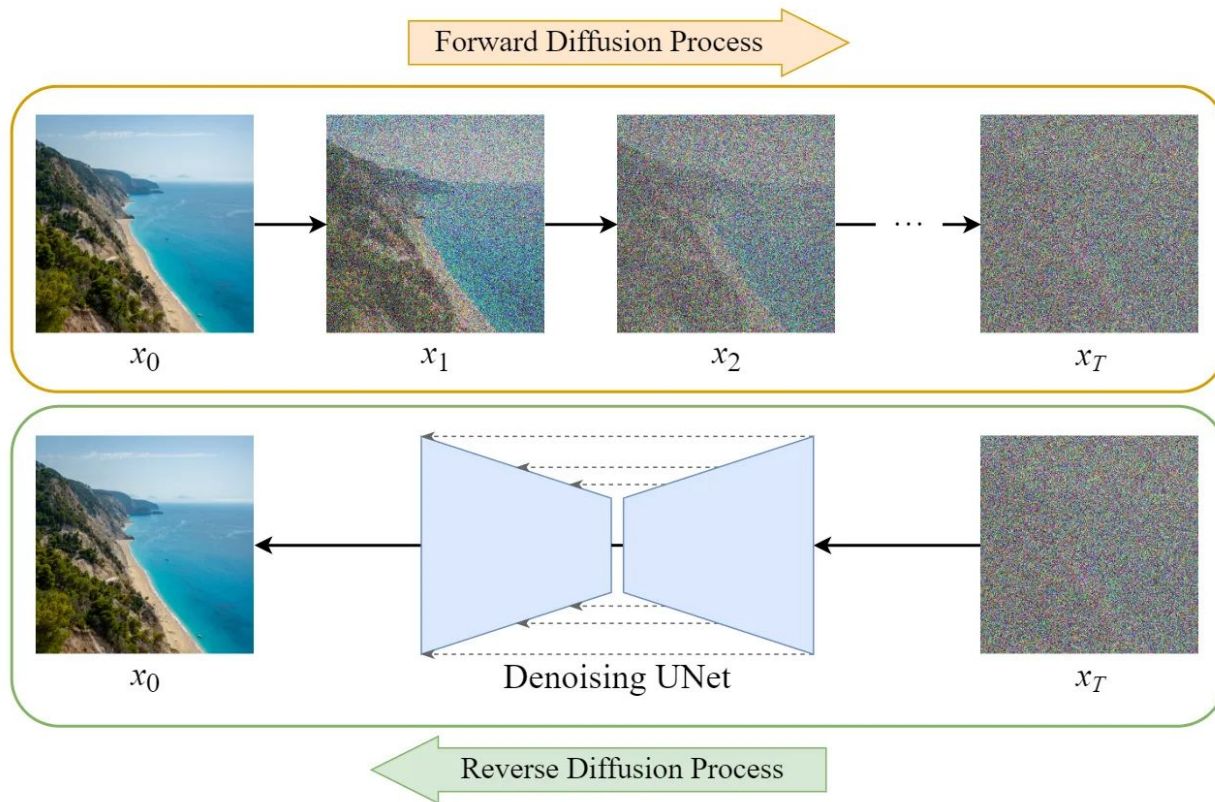
Reverse Denoising Process

Formal definition of forward and reverse processes in T steps:

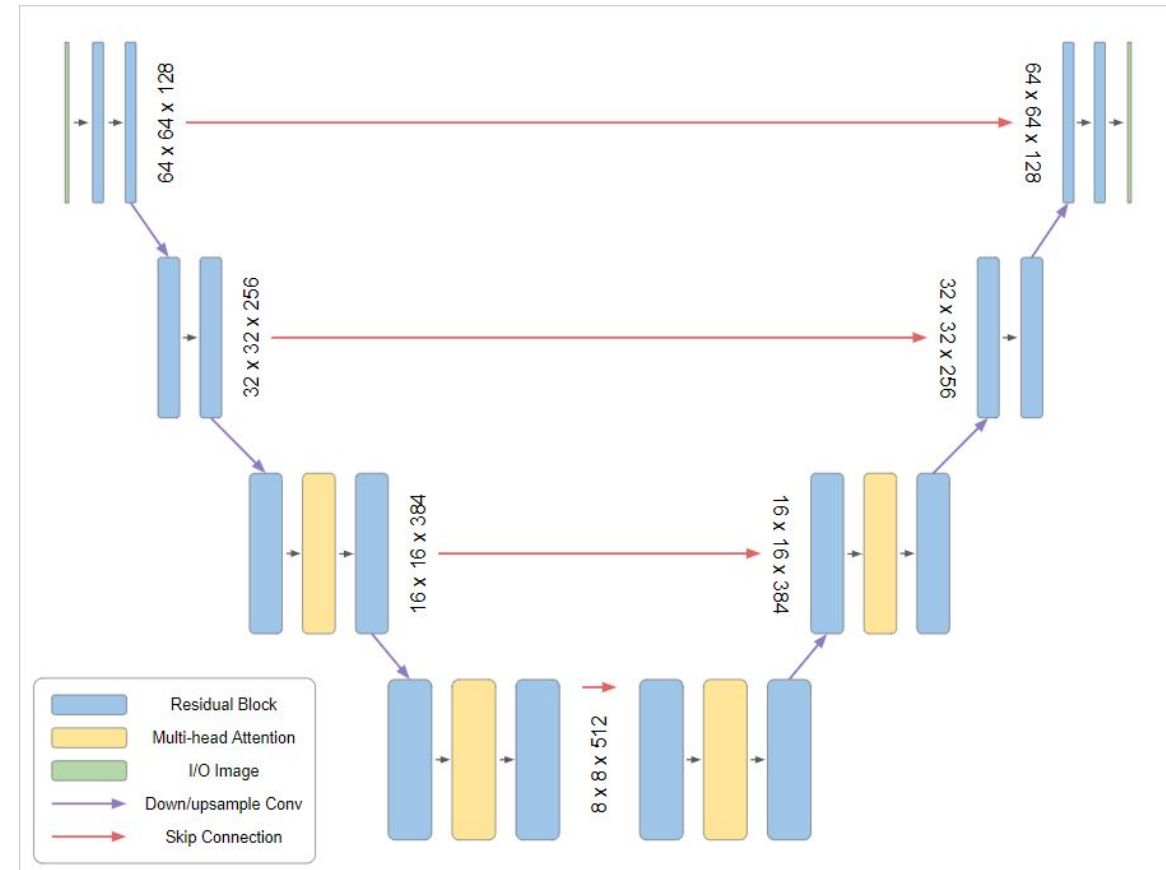


04 | Diffusion Model

Diffusion



UNet



04 | Diffusion Model

Summary

Training and Sample Generation

Algorithm 1 Training

```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
        $\nabla_{\theta} \left\| \epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2$   
6: until converged
```

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
2: for  $t = T, \dots, 1$  do  
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$   
5: end for  
6: return  $\mathbf{x}_0$ 
```

04 | Diffusion Model

Condition



04 | Diffusion Model

Condition



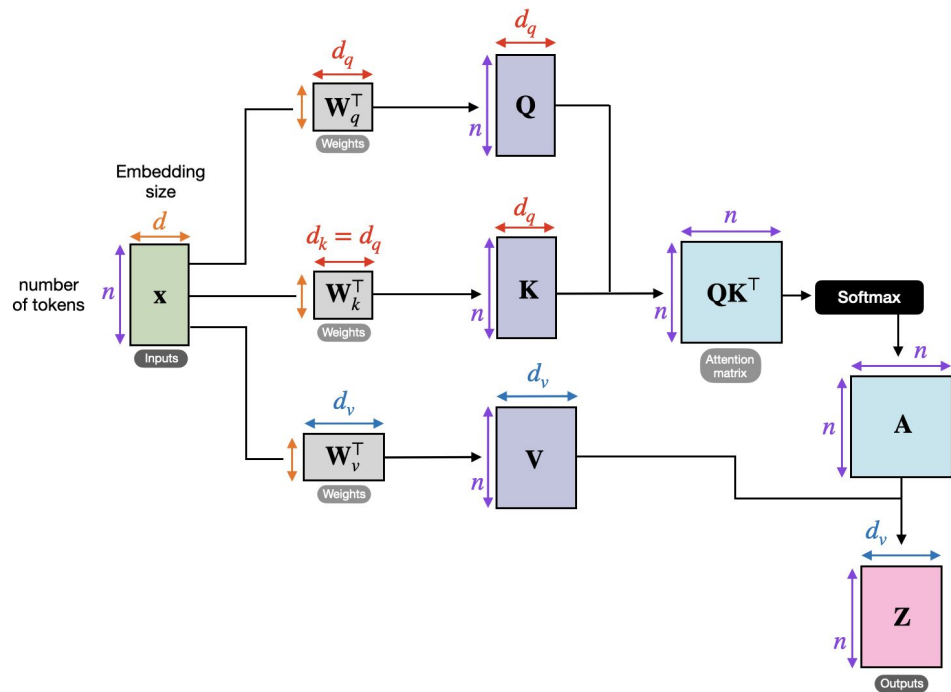
Latent Space



04 | Diffusion Model

Condition

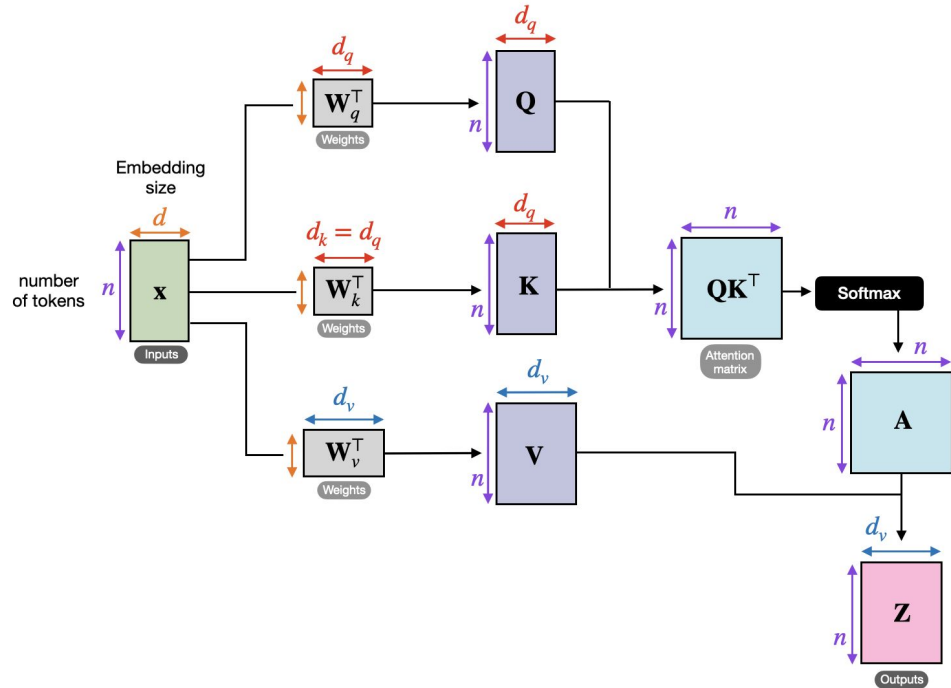
Self
Attention



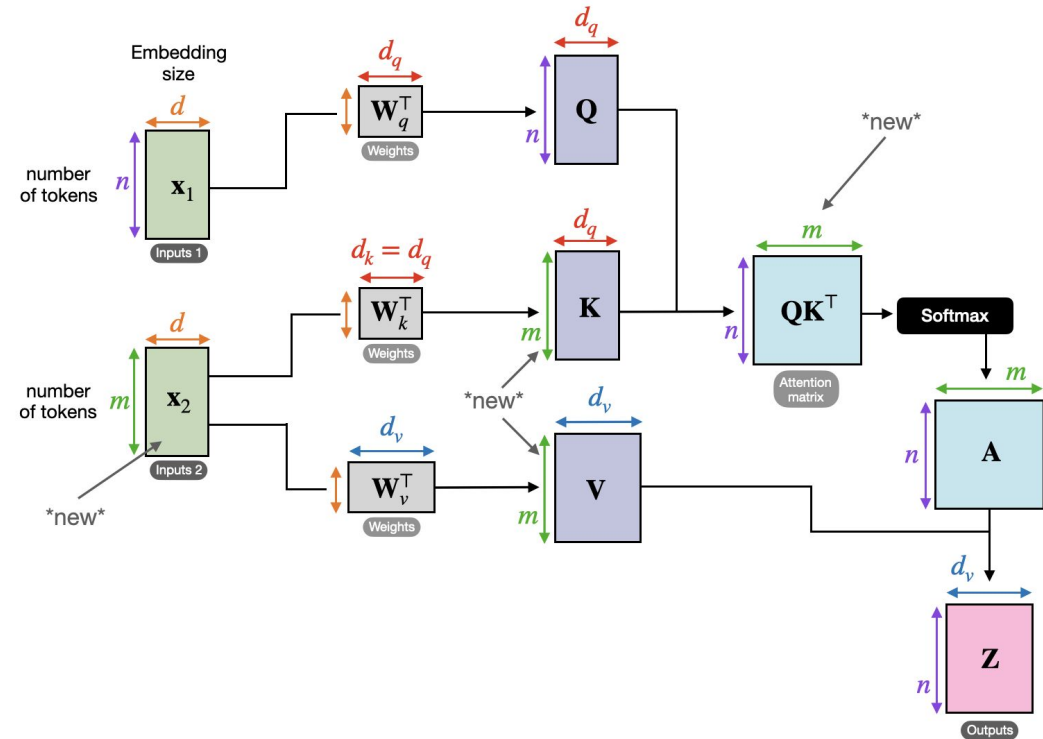
04 | Diffusion Model

Condition

Self Attention



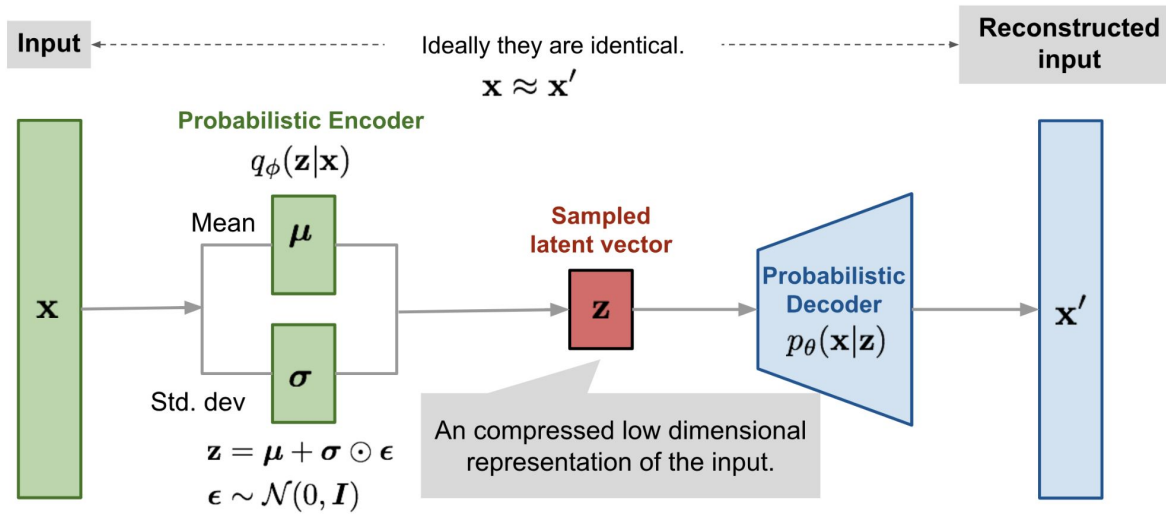
Cross Attention



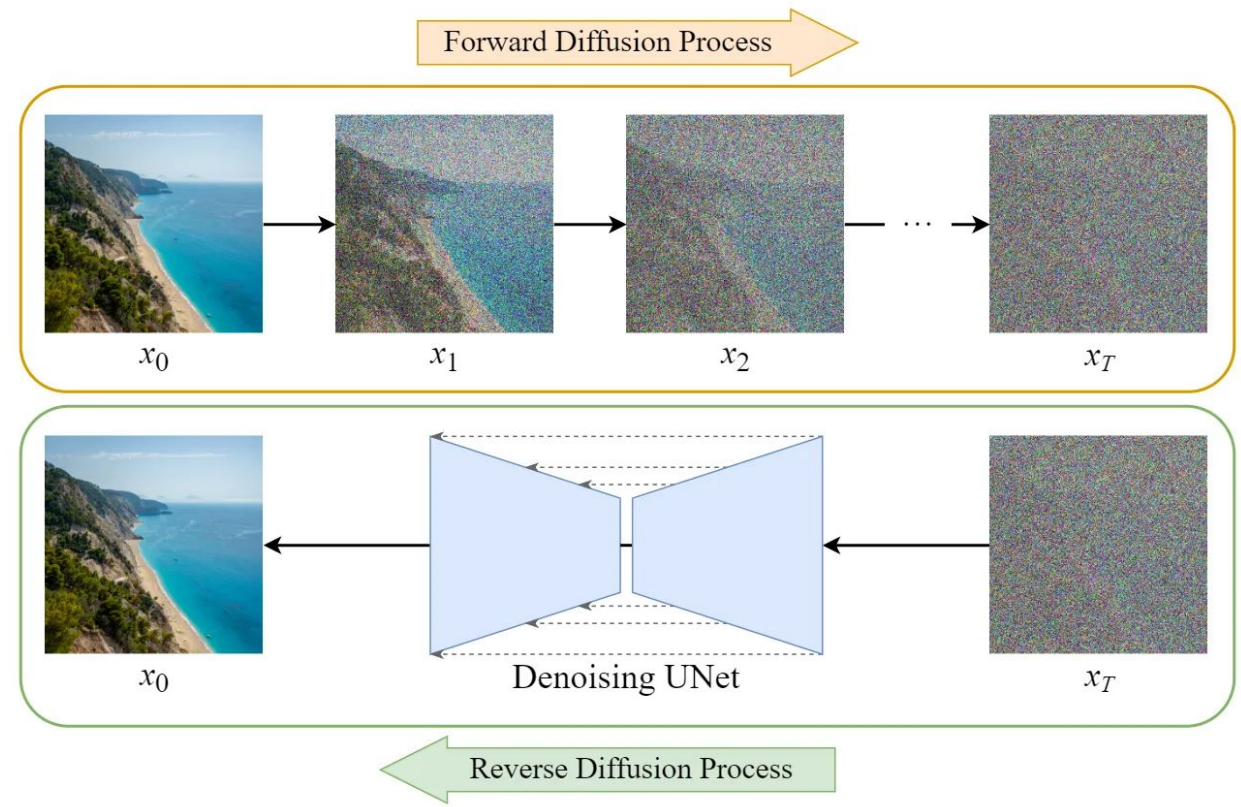
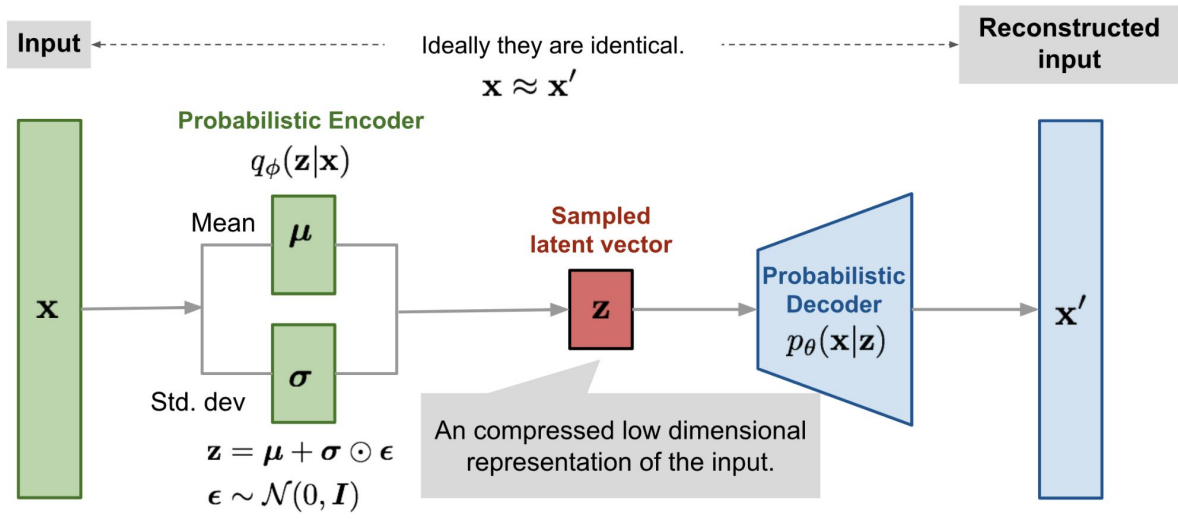
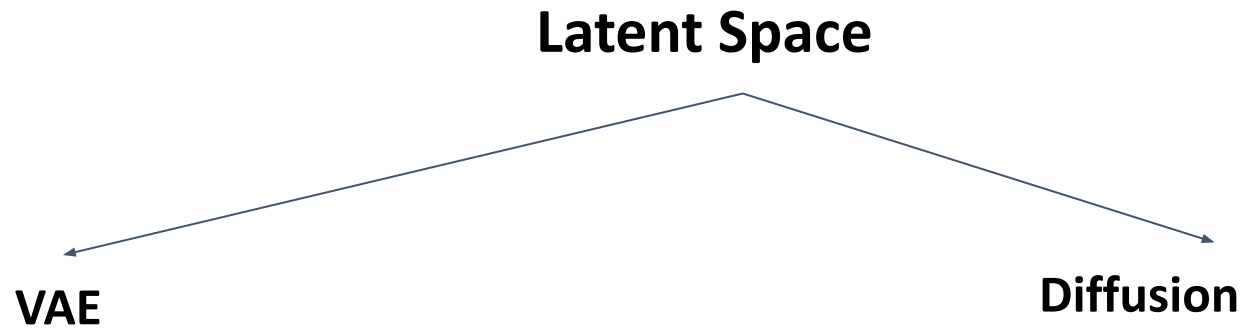
04 | Diffusion Model

Latent Space

VAE

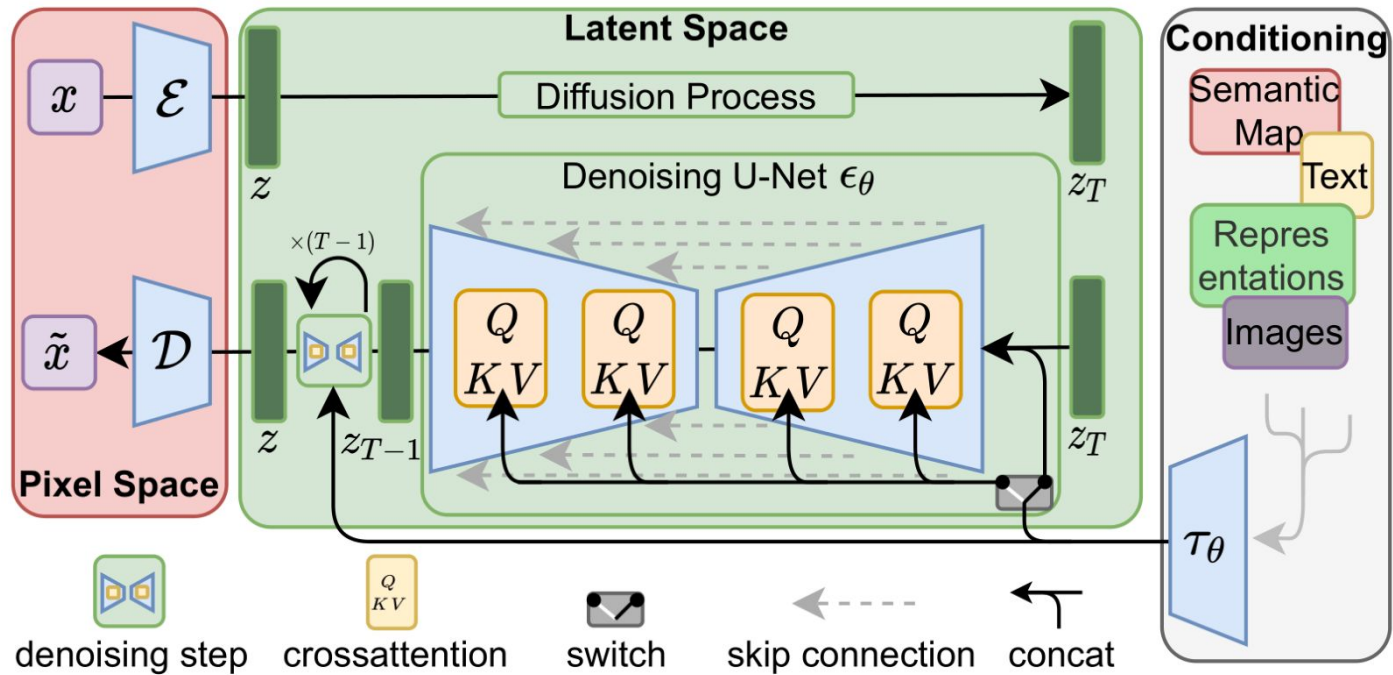


04 | Diffusion Model



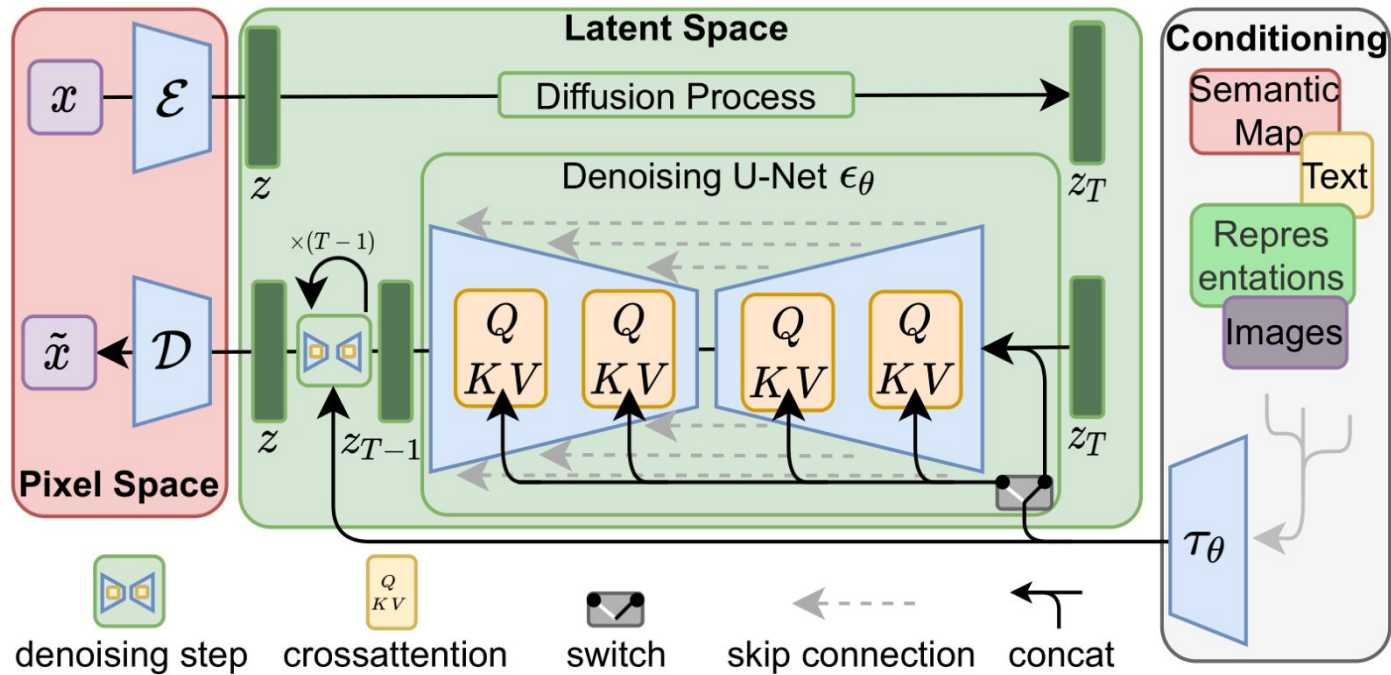
04 | Diffusion Model

Latent Diffusion Model



04 | Diffusion Model

Latent Diffusion Model



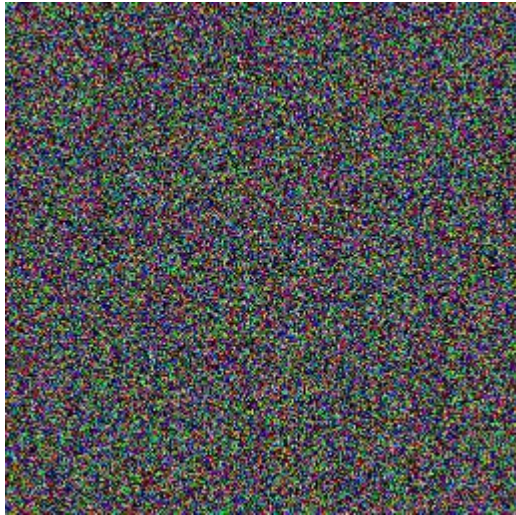
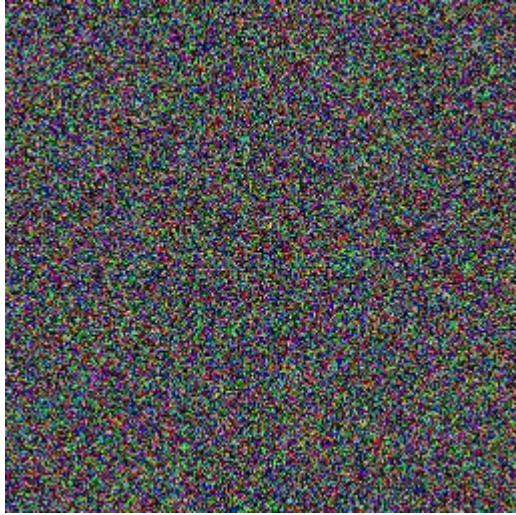
Imagen

A group of teddy bears in suit in a corporate office celebrating the birthday of their friend. There is a pizza cake on the desk.

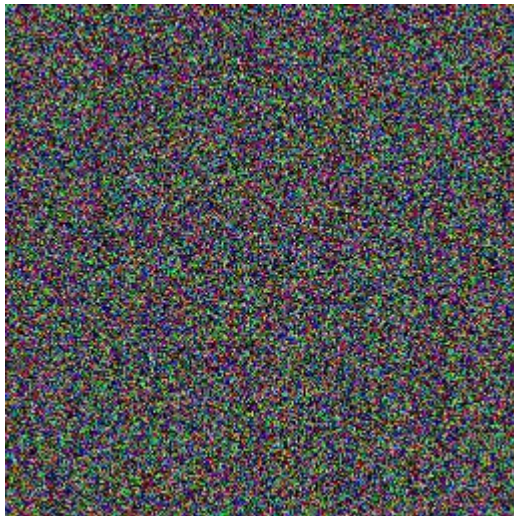
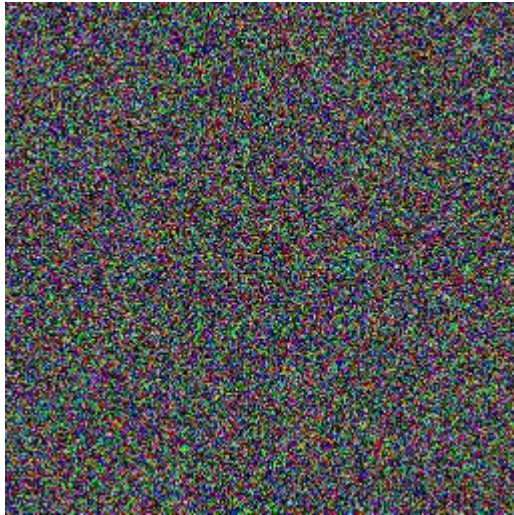


["Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding", Saharia et al., 2022](#)

05 | Demo



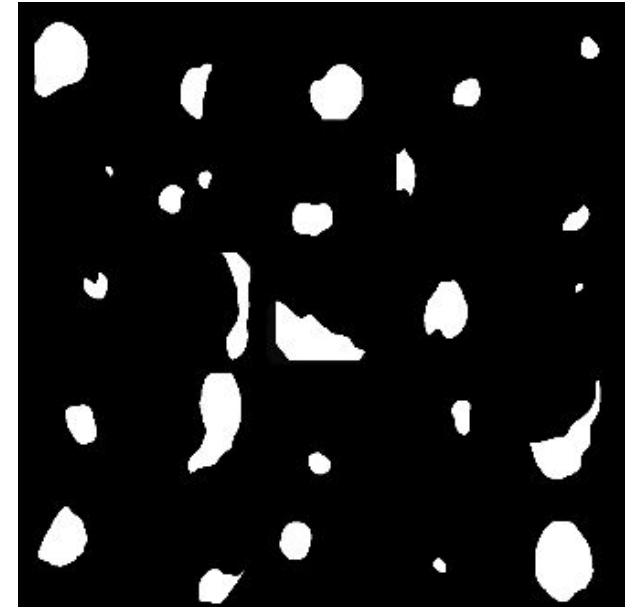
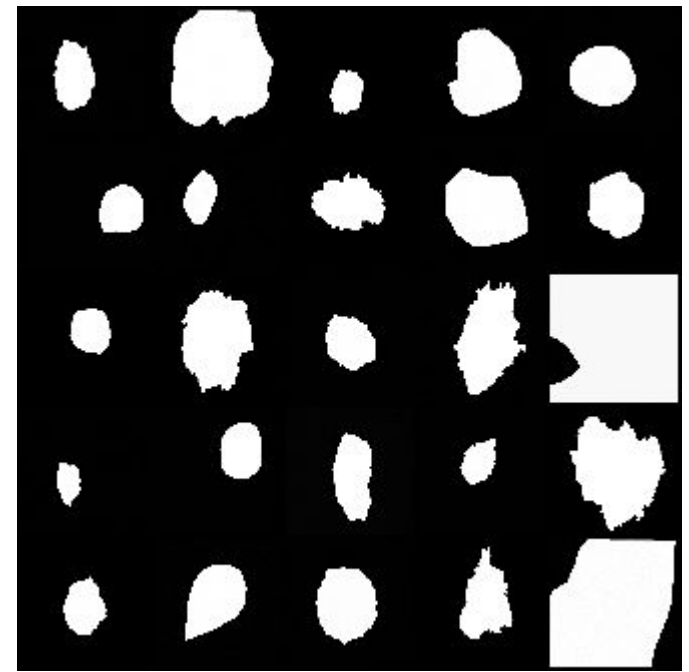
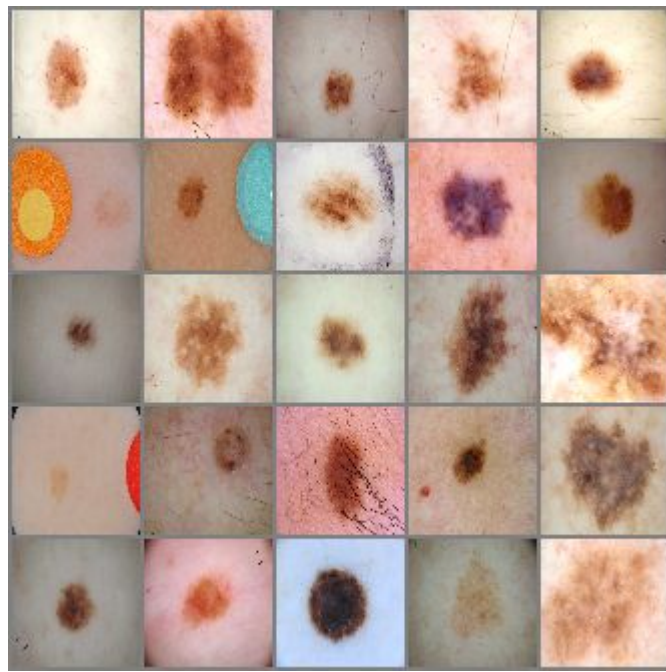
05 | Demo



05 | Demo



05 | Demo



Demo WebUI

